

Milestone 3 Report

Geographical Pattern Visualization

Mapping the Spatial Distribution of US Natural Disaster Declarations

Project Phase Milestone 3 of 4	Focus Area Geographical Pattern Analysis	Duration Weeks 5–6 (2 weeks)
--	--	--

FEMA Web Disaster Declarations Dataset | Weeks 5–6

1. Milestone Overview

Milestone 3 builds directly on the temporal insights generated in Milestone 2. While the previous phase answered when disasters occur, this milestone answers where — mapping the spatial distribution of federally declared disasters across the United States. By aggregating FEMA Web Disaster Declarations data geographically and rendering it through choropleth maps, hotspot analyses, and comparative bar charts, this milestone reveals the regional patterns and state-level concentrations that define the US disaster landscape.

2. Goals & Deliverables

2.1 Primary Goals

This milestone targets the following analytical objectives, drawn directly from the project specification:

- Aggregate disaster declaration data geographically, computing total declarations per state and counts of specific incident types per state.
- Create a choropleth map showing the total number of disaster declarations per state across the full dataset history.
- Develop maps identifying hotspots for specific incidentType categories — for example, hurricanes concentrated along Gulf and Atlantic coastlines, wildfires in western states, and tornadoes across the central plains.
- Produce comparative bar charts ranking states by total declaration counts or by frequency of selected incident types.
- Ensure all maps and charts include proper legends, meaningful color scales, and clearly labeled axes for accurate interpretation.

2.2 Minimum Deliverable Requirement

Success Threshold

Per the project plan: a minimum of 3–4 distinct geographical visualizations must be created, including at least one choropleth map and one bar chart. All visualizations must be accurately mapped to correct state-level geographies, include meaningful color scales, and clearly highlight geographic differences or concentrations of disaster activity.

3. Technical Approach

3.1 Data Aggregation

Using Pandas, the cleaned dataset from Milestone 1 is grouped and aggregated along geographic dimensions. Key operations include:

- Grouping by state to compute total disaster declaration counts per state (declarations_per_state).
- Grouping by state and incidentType to generate a state-by-type matrix enabling incident-specific choropleth maps.
- Filtering the dataset to isolate top incident categories (Flood, Hurricane, Fire, Tornado, Severe Storm) for hotspot mapping.

- Merging aggregated state data with US geographic shapefiles using GeoPandas for spatial join and map rendering.
- Sorting states by total declaration count for ranked bar chart generation.

3.2 Visualization Libraries

The following Python libraries from the project tech stack are used in this milestone:

Library	Usage in Milestone 3
Pandas	Data loading, groupby aggregation by state and incidentType, merging with geo data
GeoPandas	Spatial data handling, shapefile loading, and choropleth map rendering via plot()
Folium	Interactive choropleth maps with hover tooltips, zoom, and state-level detail popups
Matplotlib	Base rendering engine for static choropleth maps and comparative bar charts
Seaborn	High-level styling for state comparison bar charts and ranked visualizations
Plotly Express	Interactive bar charts and scatter geo maps with hover filtering and animation

3.3 Geographic Data Sources

State-level geographic boundaries are sourced from publicly available US Census Bureau TIGER/Line shapefiles. These shapefiles are loaded via GeoPandas and spatially joined with the aggregated FEMA declaration data using standard two-letter state abbreviations as the join key. This ensures accurate geographic rendering for all 50 states plus Washington D.C.

4. Expected Geographical Visualizations

The four core visualizations produced in this milestone each address a distinct facet of the spatial disaster distribution. Together they form a comprehensive geographical portrait of FEMA-declared disasters across US history.

4.1 Total Disaster Declarations Per State (Choropleth Map)

A full US choropleth map where each state is shaded according to its total number of FEMA disaster declarations across the complete dataset. A sequential color scale (light-to-dark) encodes declaration frequency, with darker shades indicating higher disaster activity. States with the most declarations — such as Texas, California, and Oklahoma — are expected to appear most prominently.

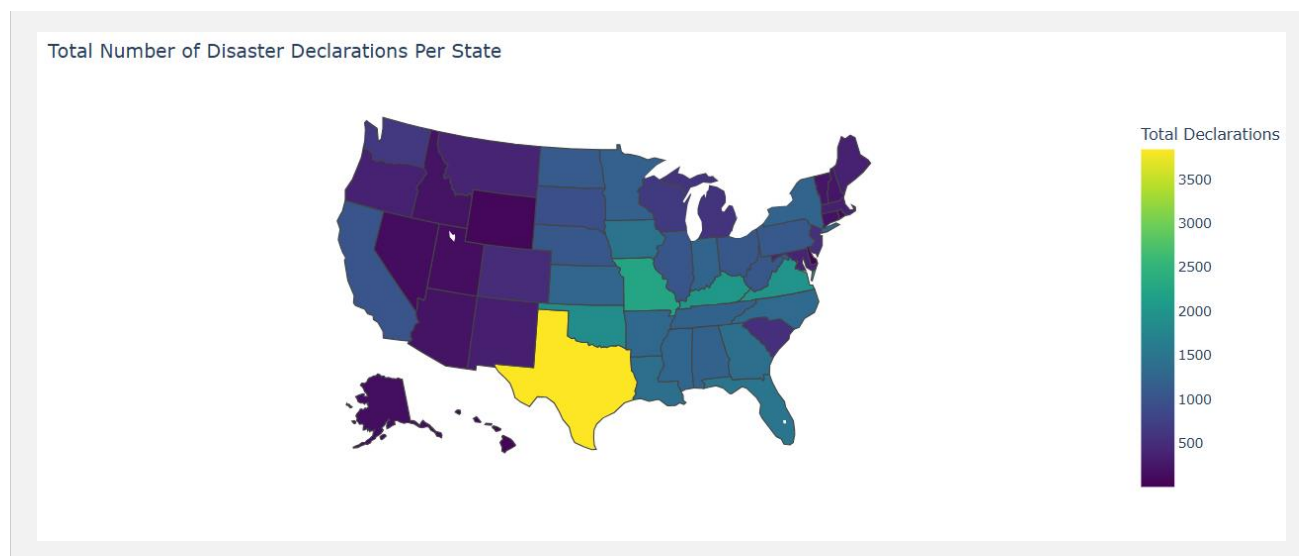


Figure 1. Choropleth map of total FEMA disaster declarations by state. Southern, Gulf Coast, and Plains states are expected to show the highest declaration concentrations due to hurricane, tornado, and flood exposure.

4.2 Incident Type Hotspot Maps

A series of incident-specific choropleth maps, each displaying the concentration of a single disaster type across states. This allows direct comparison of the geographic footprint of each major disaster category:

- Hurricane hotspots concentrated along the Gulf Coast (Texas, Louisiana, Florida) and Atlantic coastlines.
- Wildfire hotspots in western states, particularly California, Oregon, Washington, and Arizona.
- Tornado activity concentrated in the central plains and Midwest — the traditional Tornado Alley (Oklahoma, Kansas, Nebraska, Texas).
- Flood declarations broadly distributed but concentrated in Mississippi River basin states and the Southeast.

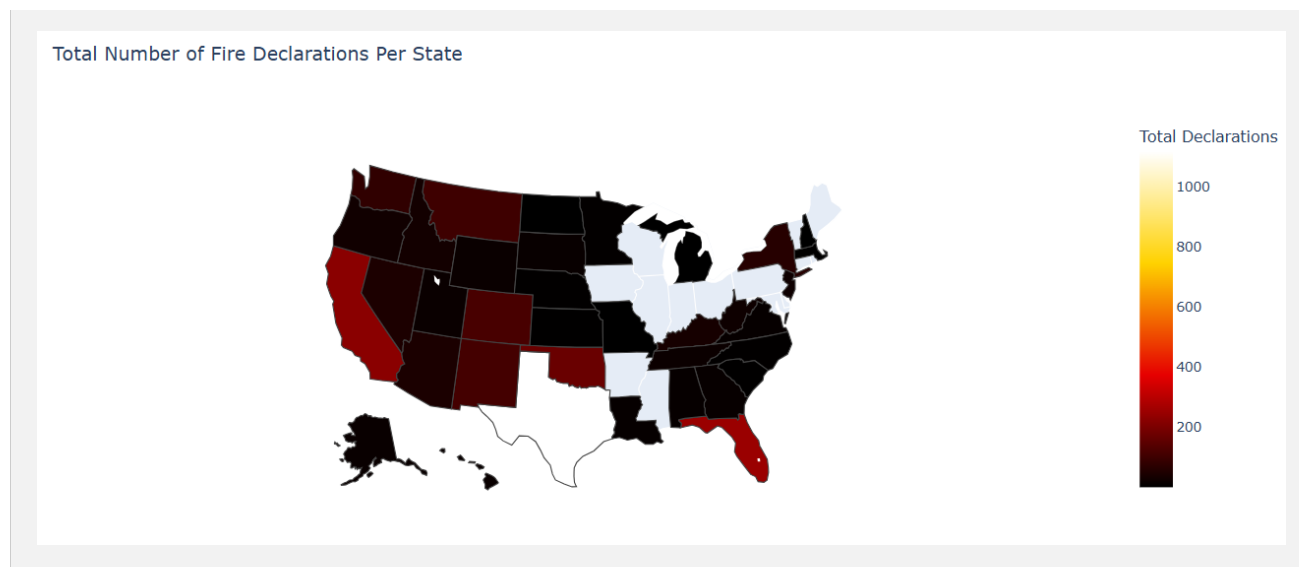


Figure 2. Panel of incident-specific choropleth maps revealing distinct geographic footprints for major disaster types. Each map uses a consistent color scale to enable cross-panel comparisons.

4.3 State Ranking Bar Chart — Total Declaration Counts

A horizontal bar chart ranking all US states from highest to lowest total FEMA disaster declarations. This chart provides a precise, quantitative complement to the choropleth map — making exact counts readable and enabling direct state-to-state comparison.

- Bar colors are mapped to geographic region (Southeast, Midwest, West, Northeast) for additional spatial context.
- An interactive Plotly version supports hover detail showing exact declaration counts and percentage of national total.
- Top 20 states are highlighted in a separate zoomed view for detail legibility.

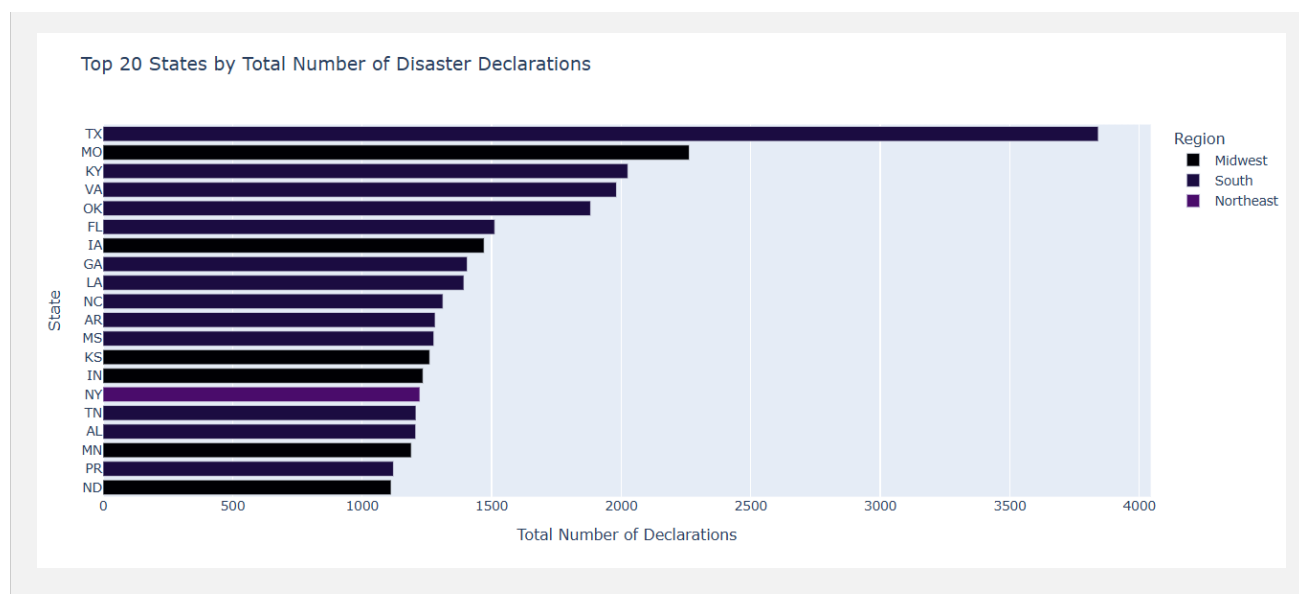


Figure 3. State ranking bar chart by total FEMA declarations. Texas, California, and Florida are anticipated to lead due to their size, population density, and diverse natural hazard exposure.

4.4 State-by-Incident Type Heatmap

A two-dimensional heatmap with states on one axis and major incident types on the other, with cell color intensity representing declaration frequency. This visualization simultaneously reveals which states are most active overall and which disaster types dominate each region, supporting multi-dimensional geographic comparisons in a single chart.

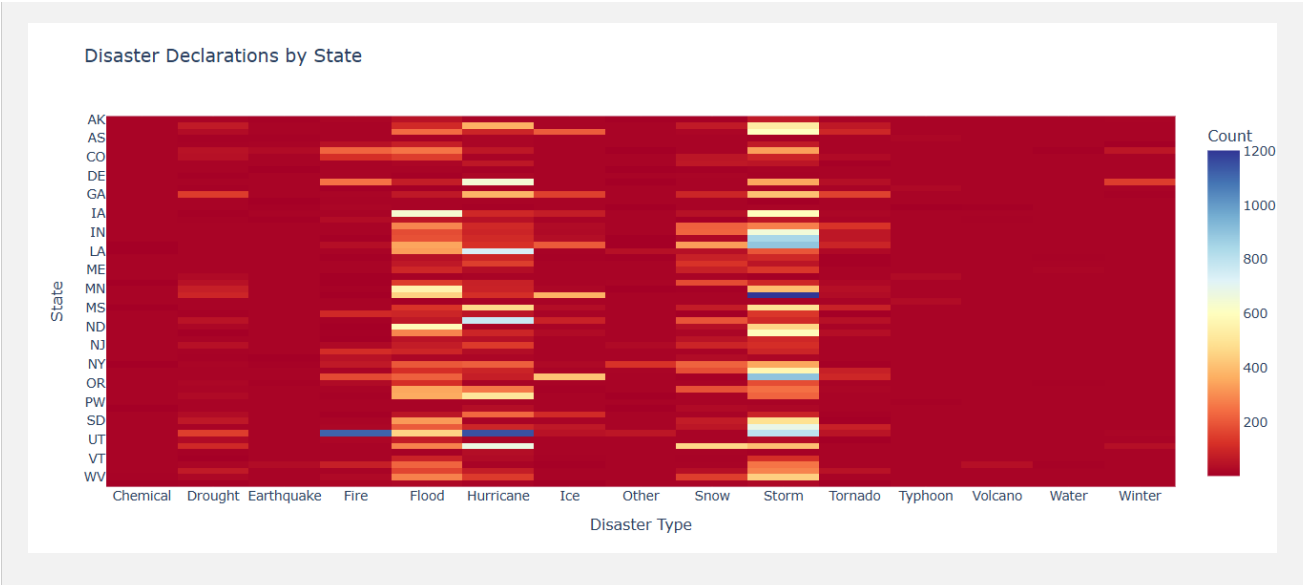


Figure 4. State-by-incident-type heatmap enabling simultaneous geographic and categorical analysis. Florida's hurricane dominance and California's fire activity are expected to produce distinct high-intensity cells.

5. Evaluation Plan

Per the project milestone plan, the following evaluation criteria are applied at the end of Week 6 to assess successful completion of Milestone 3:

Metric	Target	Evaluation Method
Map Accuracy	Data correctly mapped to all 50 states with no geographic misattribution	Cross-reference 5 states against raw Pandas groupby output
Visual Clarity	Color scale meaningful; legend present; state boundaries clearly distinguishable at standard resolution	Peer review + visual inspection at 100% zoom
Spatial Effectiveness	Visualizations clearly highlight geographic differences and regional disaster concentrations	Reviewer assessment: can a reader identify the top 5 states without reading labels?
Label Completeness	All maps include title, legend, color scale bar, and data source citation	Checklist: title, legend, scale, source on every visual
Deliverable Count	Minimum 3–4 distinct geographical visualizations produced	Count of final outputs in Jupyter Notebook

5.1 Success Criteria Summary

- At least 3–4 distinct geographical visualizations are finalized and included in the Jupyter Notebook.
- The choropleth map accurately reflects total declaration counts per state with a clear, meaningful color scale.
- Hotspot maps for specific incident types reveal expected geographic patterns (coastal hurricanes, western fires, plains tornadoes).
- State ranking bar charts enable precise quantitative comparison across all 50 states.
- One or more visualizations include an interactive Folium or Plotly version for dynamic zoom and hover exploration.

6. Workflow Pipeline

The Milestone 3 workflow follows the same structured pipeline defined across the project, extending the data transformation approach into the spatial visualization domain:

Step	Action
Step 1	Load cleaned Milestone 1 dataset into Pandas DataFrame. Verify state codes and incident type columns.
Step 2	Aggregate data by state using <code>groupby()</code> to compute <code>total_declarations_per_state</code> and per-type counts.
Step 3	Load US Census Bureau state shapefiles into GeoPandas. Perform spatial join with aggregated state data.
Step 4	Generate Figure 1 — Total declarations choropleth map using GeoPandas <code>plot()</code> with sequential color scale.
Step 5	Generate Figure 2 — Incident type hotspot panel maps (Hurricane, Fire, Tornado, Flood) using filtered subsets.
Step 6	Generate Figure 3 — State ranking horizontal bar chart using Matplotlib/Seaborn with regional color coding.
Step 7	Generate Figure 4 — State × Incident Type heatmap using Seaborn with annotated cell values.
Step 8	Build interactive versions of choropleth maps using Folium with GeoJSON overlays and tooltip popups.
Step 9	Review all maps for labeling completeness, geographic accuracy, and interpretability.
Step 10	Document findings and annotate Jupyter Notebook with interpretive commentary on regional patterns.

7. Conclusion

Milestone 3 transitions the project from temporal to spatial analysis, completing the two primary dimensions of disaster pattern investigation. By applying GeoPandas for geographic data handling, Folium for interactive mapping, and Matplotlib and Seaborn for static charts, this milestone produces a suite of

geographical visualizations that reveal where US natural disasters are most concentrated — and which types dominate each region.

The methodology mirrors the structured pipeline of the DeepVision project: just as DeepVision generated spatial density maps that located crowd concentrations within image frames, Milestone 3 generates geographic density maps that locate disaster concentrations across the US landscape. Each visualization answers a specific analytical question about where FEMA-declared disasters cluster, providing the spatial foundation for the project's final synthesis and recommendations.